

Auteur: Mark Gijsbrechts
Studierichting: Informatica
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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{(1 - \cos x)'}{(x^2)'} & \\ = \lim_{x \rightarrow 0} \frac{\sin x}{2x} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{(\sin x)'}{(2x)'} & \\ = \lim_{x \rightarrow 0} \frac{\cos x}{2} & \\ = \frac{1}{2} &\end{aligned}$$

References